NLP

→ Language is a primary means of communication used by humans & tool to express the greater part of our ideas and emotions.

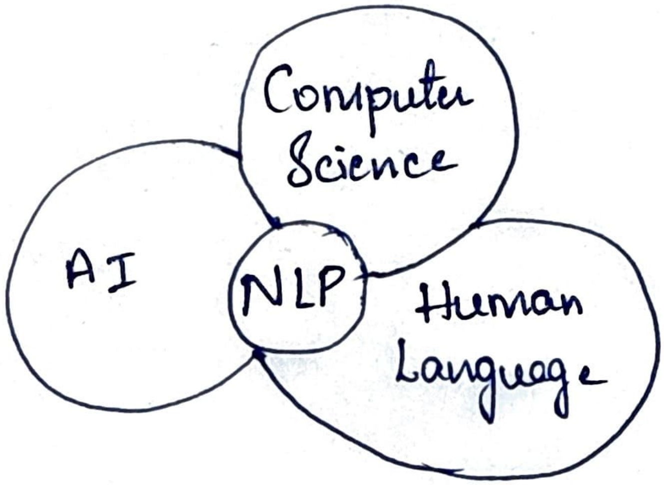
→ It ~~&~~ shapes ~~thought~~ thought, content helps represent the language in real-time

→ NLP is concerned with development of computational models of aspects of human language processing; there are 2 main reasons:-

1. To develop automated tools for language processing
2. To gain a better understanding of human communication.

→ Building computational models with human language processing abilities requires a knowledge of how human acquires, store & process language.

NLP is a part of Computer Science, Human language and AI.



It outlines 2 key philosophical approaches to understandi
language or knowledge acquisition

→ Historically there are 2 major Approaches to NLP :-

1. Rationalist Approach
2. ~~Empirical (or)~~ Empiricist Approach

→ Rationalist Approach :-

* It is a Early approach

innate(inbort ability feature)

* It assumes the existence of some language faculty in the human brain.
* Few Supporters of this approach argue that it is not possible to learn a complex thing like natural language from limited sensory input.

→ Empiricist Approach

* It Don't believe in existance of a language faculty.
* It Believes in the existence of some general organization principles such as pattern recognition generalization & association.

* Learning of detailed structures takes place through the application of these principles on sensory inputs available to the child.

⇒ CHALLENGES of NLP :-

1) One Word, Many Meanings :-

Some words have more than one meaning. The computer must decide which meaning is correct.

Ex:- The word "bat" can mean a flying ~~bat~~ animal (or) cricket bat.

NLP must know which one we are talking about.

2) Grammer Rules Changing meaning :-

If we change the order of words, the meaning also changes.

Ex:- "He is eating" → Normal Sentence

"Is he eating". → Question.

NLP must know grammer rules to find correct meaning

3) Slang and Short forms :-

People use short forms and slang words, especially in chats or messages.

Ex:- "u" means "You"

"idk" means "I don't know"

Computers cannot always understand these easily.

4) Need Context (Situation)

Some sentences need extra information to understand properly this is called context.

EX:- "It is Hot"

is it about the weather? Teo? (or) Room Temp?

NLP must understand the situation to give the right meaning.

5) Different Language and Accents

People speak many languages like. English, Telugu. Kannada, Hindi etc. Even in ~~this~~ the same language, accents and pronounciations are different.

EX:- "Color" (English) vs "colour" (British)

→ Kannada spoken in Bangalore ~~and~~ vs uttara Karnataka has small changes NLP must handle all these differences.

→

6) No Emotions in Machines :-

Humans can understand feelings and emotions in words. But computers don't have emotions, So they can't understand tone or mood.

EX:- If someone says "I'm fine..." (in Sad tone) a human understands sadness. But a computer May think the person is happy. Where understanding emotions is very hard for NLP.

⇒ Language & Grammer

* Language is a means of communication to share knowledge & expressions.
* Grammer is a set of rules for constructing sentences in a language. used to analyze & understand the structure in text data.

Ex:- H2H intraction if we use english. Then we have to follow some grammer rules.

→

Why Grammer is Important?

-> ~~A~~ for Automatic processing of language, the Grammer rules and exceptions of a language is to be explained to the computer.

-> Grammer defines language by specifying how sentences are structured.

-> It helps computer to parse and generate sentences accurately.

-> These rules relate information to coding devices at the language processing level and not at the world ~~language~~ level.

Challenges in NLP Grammer?

-> Natural languages are constantly changing.
Ex:- The word "awesome" originally ~~meant~~ "filling with awe" but now is commonly used to mean "great" or "fantastic".

→ There are many hard-to-specify exceptions.

EX:- In English, the plural for "mouse" is "mice", but the plural for "House" is "houses" rather than "hice".

→ Several efforts have led to various types of grammers. Some of these grammars focus on derivation while other focus on relationships.

Types of Grammer :-

1→ Transformational Grammer
2→ Lexical Grammer
3→ Government and Binding Grammer
4→ Generalized Phrase Structure Grammer
5→ Dependency Grammer.
6→ Paninian Grammer
7→ Tree-Adjoining Grammer.

→

Grammer Type	Definition	Focus	Key Concept
1) Transformational Grammer	Describes how different sentence forms are derived through transformations.	Sentence structure transformation.	Sentence derivation through rules.
2) Lexical Functional Grammer (LFG)	Separates syntactic structure from functional aspects.	Syntax - meaning correspond-ence	Syntactic and Semantic mapping
3) Government & Binding	Defines syntactic structures based on hierarchical rules.	Constraints in Syntax	Phrase and Constraint relationships.
4) Generalized Phrase Structure Grammer	Formal grammer specifying hierarchical sentence structures	Phrase Structure rules	Rule-based Structure formation.
5) Dependency Grammer	Emphasizes relationships between words rather than constituent structure.	Dependency between words	Head-dependent word pairs.
6) Paninian Grammer	Derived from ancient Sanskrit grammer, emphasizing morphological structures.	Rule-based language representation	Morphological richness and structural rules.
7) Tree-Adjoining Grammer	Uses ~~to~~ tree structures to model sentence formation	~~Tree~~ Tree-based hierarchical structure	Recursive sentence modeling.

After all those 7 grammers one more was introduced → Generative Grammer.

Generative Grammer :-

→ Introduced by Noam Chomsky.

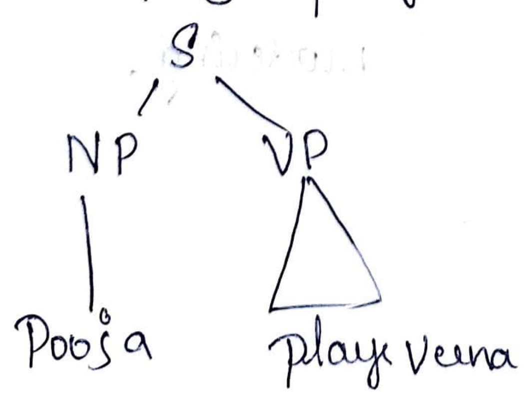
→ It uses a set of rules to specify or generate well-formed sentences.

→ It was introduced to address limitations of phrase structure grammers.

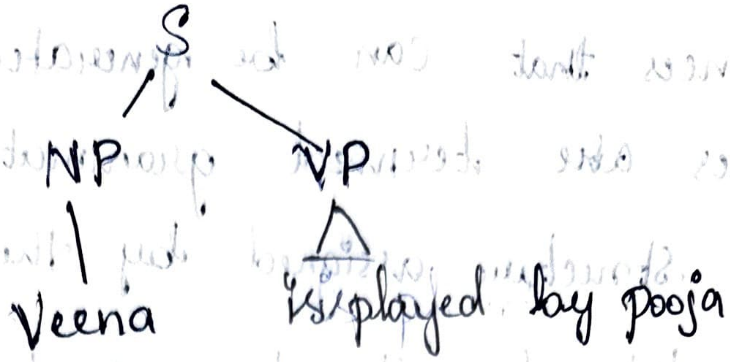
→ Each sentence in language has 2 levels of representations:-

1) Deep Structure :- Represents Meaning
2) Surface Structure :- Represents Sentence form.

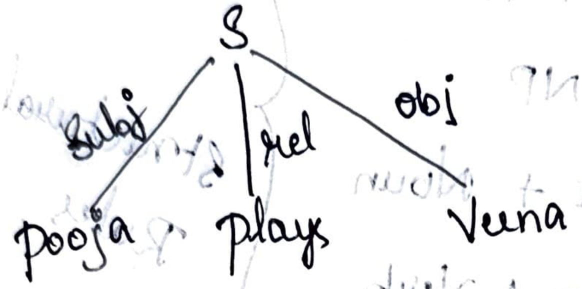
Ex:- "Pooja plays Veena"



"Veena is played by Pooja"



(Fig) Surfase Structure



(fig) Deep structure

→ The mapping from deep structure to surface structure is called out by transformations.

⇒ Components of Transformational Grammer :-

It has three Components :-

1) Phrase Structure Grammer
2) Transformation Rules
3) Morphophonemic Rules

1) Phrase Structure Grammer :-

* It consists of set of rules that generates natural language sentences and assigns structural descriptions.

→

* Sentences that can be generated using these rules are termed grammatical.

The structure assigned by the grammar is a Constituent structure analysis of the sentence.

Ex:-

S → NP + VP

VP → V + NP

NP → Det + Noun

V → Aux + Verb

Structural Rules

meaning ⇒ S = Sentence

NP = Noun Phrase

VP = Verb phrase

V = Verb

Det = Determination

Aux = Auxilary

Eg for:- Det = The, a, an ---

Verb = catch, write, eat ---

Noun = Police, Snatcher ---

Aux = will, is, can ---

2) Transformational Grammer Rules

-> These contains set of transformation rules, which transform underlying phrase structures into derived ones.

-> These rules are used to transform one surface representation to another.

Rules:-

$NP_1 - Aux - V - NP_2 \rightarrow NP_2 - Aux + be + en - V - by + NP_1$

It tells:-

① NP - Aux - V - NP

↓

② NP - Aux + be + en - V - by + NP.

It tells that the input which is having the structure can be transformed into derived one by adding some strings "be" and "en" and certain rearrangements of constituents of sentence.

Ex:- Transforming active to passive sentences:-

"The police will catch the snatcher"

↓

"The snatcher will be caught by the police"

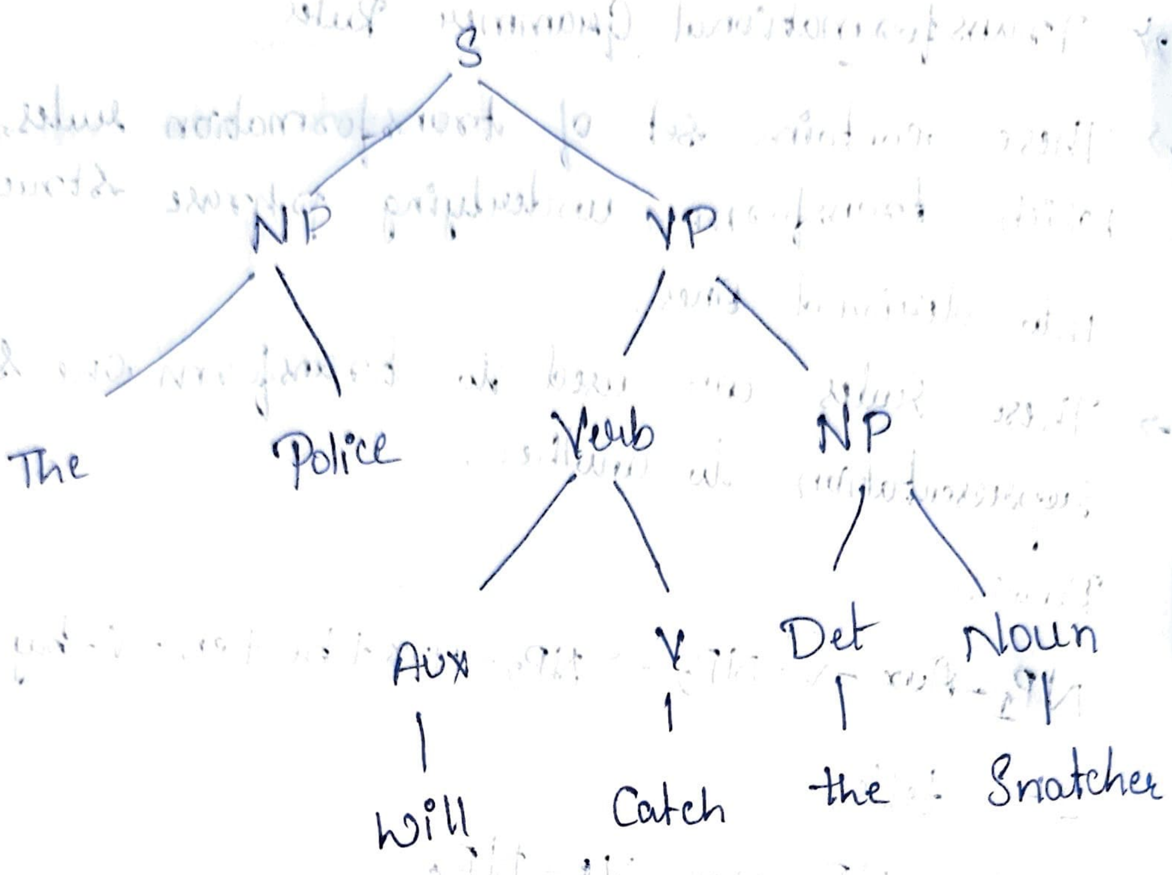


Fig:- Parse Structure of Sentence.

The passive transformation rules will convert the sentence into -

The+ Culprit + will + be + en + catch + by + police

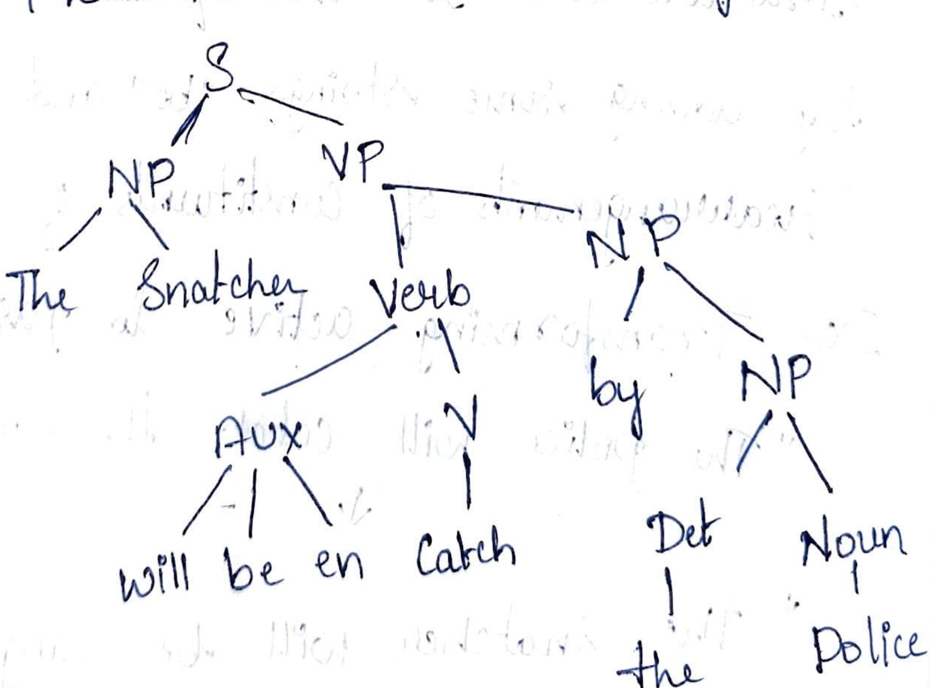


Fig:- Structure of sentence after applying passive transformation

3) Morphophonemic Rules :-

* This rules don't express conceptual categories rather they simply specify pronounciation of morphemes in context.
* Interface between phonology & morphology.

Ex:- Changes in "Sleep" and "Slept", "bind" and "bound".

'catch' + 'en' = 'caught'

=> Processing Indian Languages :-

* There are number of differences b/w Indian languages and English.
* Indian languages are very different from English. Because of this, it is difficult to process them in the same way as English.

-> Key Differences :-

1) English, which has a linear & uniform structure. Indian languages use non-linear Indic structure.

2) Indian language have SOV (Subject-Object-Verb) as default setence structure and where as English uses SVO (Subject-Verb-Object).

Eg:- She is reading book → (SVO)

Kannada → ಅವಳು. ಪುಸ್ತಕವನ್ನು ಓದುತ್ತಿದ್ದಾಳೆ → (SOV)

She Book Read

3) Free word order: words can move freely without changing meanings, this is not the case in English. "ಪುಸ್ತಕವನ್ನು ಅವಳು ಓದುತ್ತಿದ್ದಾಳೆ".

4) Spelling standardization is more difficult in Hindi than English.

5) Indian Languages have a richer set Rich set of morphological variants (~~from~~ form of words). Compared to English.

6) IL make extensive use of complex predic-ates (CPs), which involves auxiliary verbs or additional components.

(~~changing~~ main verb to additional words)

EX:- is doing

7) IL use of post-position case markers instead of pre positions, which is a syntactic difference

8) verb complexes consist of sequences of verbs. The auxiliary verbs in this sequence provide information about tense, mood & aspect.

9) Hindi & Urdu Relationship:-
- → Both share similar phonology, morphology & syntax
- → Both are free-word-order languages & use post-positions.
- → Main difference arise mainly in vocabulary:-
  - Hindi borrows from Sanskrit
  - Urdu borrows from Arabic

⇒ NLP Applications:-

1) Machine Translation
2) Speech Recognition
3) Speech Synthesis
4) Information Retrieval
5) Information Extraction
6) Question Answering
7) Text Summarization
8) Natural language Interfaces to Databases.

1) Machine ~~Techn~~ Translation :-

* It automatically translate text from one human language to another

* To do that, it is necessary to have an understanding of words and phrases, grammers of the 2 languages involved, semantics of the languages and world knowledge.

2) Speech Recognition :-

* It is a process of mapping acoustic speech signals to a set of words.

* The difficulties arise due to wide variations in the pronounciation of words, homonym (Ex:- dear and deer) and acoustic ambiguities (eg:- in the rest and interest).

3) Speech Synthesis :-

* It refers to automatic production of speech.

* Such systems can read out your mails on telephone, or even read out a storybook for you.

4) Natural language Interfaces to Databases :-

It allow querying a structured database using natural language sentences.

5) Information Retrieval:-

* This is concerned with identifying documents relevant to a user's query (or) Question.
* NLP have found useful applications in information retrieval. ~~In~~
* Indexing, query modification, and knowledge bases have also been used in IR system to enhance performance.

Eg:- By providing methods for query expansion.

6) Information Extraction :-

* The System captures and outputs factual information contained withing a document.
* Similar to an information retrieval system, it responds to a users information need.
* But, it unlike in an information retrieval system, the information need is not expressed as a keyword query.
* Instead it is specified as pre-defined database schemas or templates.

7) Question Answering:-

* Given a question and a set of documents, a question answering system attempts to find to precise answer, or at least the precise portion of text in which the answer appears.

8) Text Summarization:-

This deals with the creation of summaries of documents and involves syntactic, semantic and discourse level processing of text.

=> INFORMATION RETRIEVAL

Introduction:-

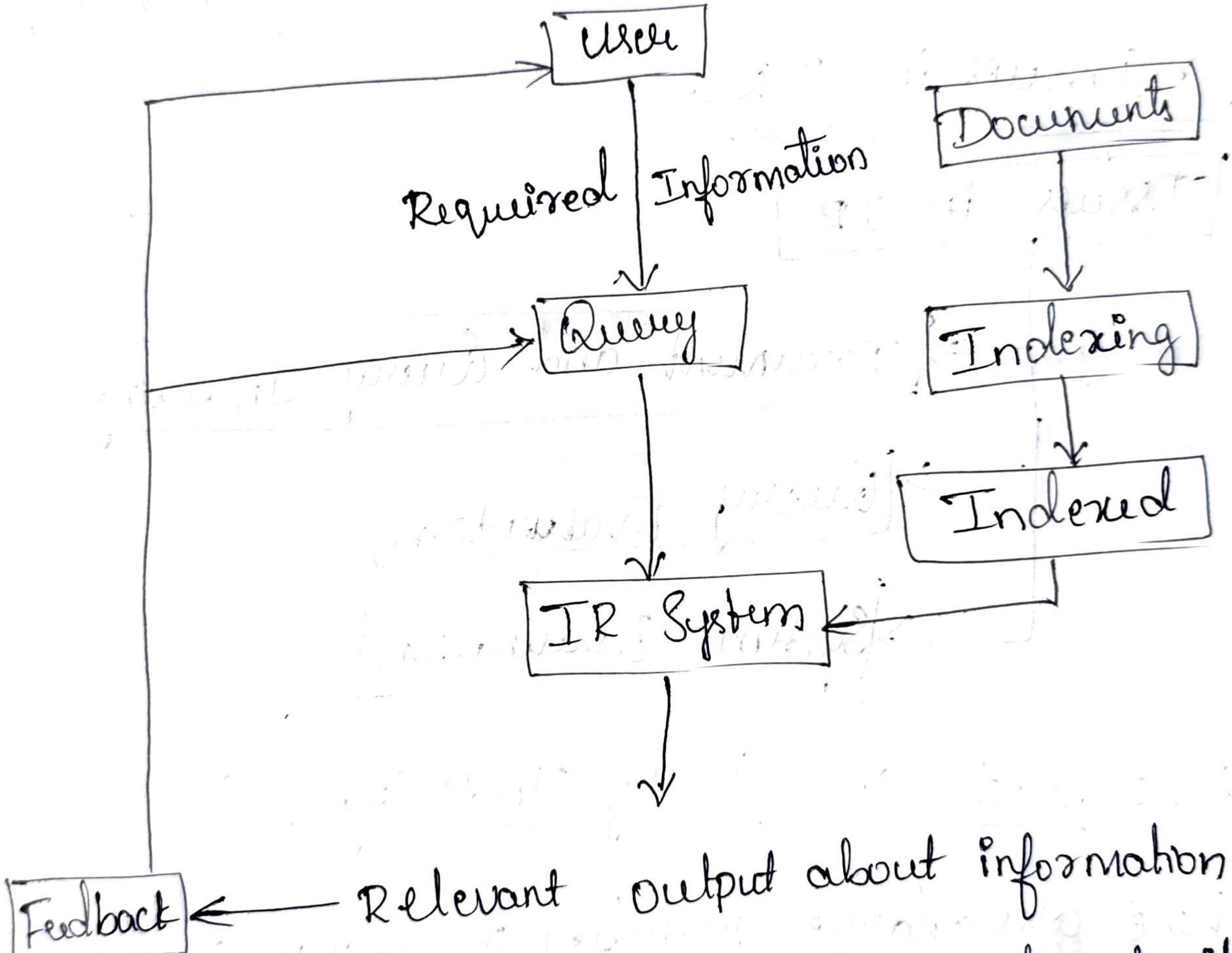
* Information Retrieval is a software program that deals with the organization, stores, retrieves and evaluates information from document repositories, particularly textual data.
* The system assists users in finding relevant information they require but it does not explicitly provide direct answers.

Purpose of IR Systems:-

* IR systems help identify and locate documents that might contain the required information.

* Documents that satisfy user requirements are called relevant documents.
* A perfect IR system retrieves only relevant documents.

How IR System works?



1) A user requires information and submit a query to system in natural language.
-> The system contains documents that need to be organized.
-> These documents go through an indexing

process to create a structured format for faster searching.

* → Once indexed, they are stored in indexed document Database.

2) The IR system processes the user query.

3) Relevant documents are retrieved as the o/p.

Major Issues in IR :-

Issues in IR

- → Document and Query Indexing
- → Query Evaluation
- → System Evaluation

(1) Document and Query Indexing :-

* There are a large number of documents, and Organizing them efficiently is challenging.
* If the indexing process is not efficient, retrieving relevant information quickly becomes difficult.

* Factors to consider: Accuracy, exhaustiveness, and manipulation by computer.
* Indexes are used to make search operations faster.
* Efficient indexing is required for fast and accurate searches.
  EX:- Library, Data base.

② Query Evaluation :-

* The system must ensure that the user gets the most relevant results for their query.
* When there is a lot of information available, filtering the right content is a challenge.

challenges include:-
- Uncertainity - Representation may not reflect true semantics.
- Vagueness : user queries may lack clarity.
* The system must ensure that users recive the most relevant results.
  EX:- Smart phones should only show Relevent data not any random information.

(3) System Evaluation :-

* The IR System needs to be evaluated to measure how efficiently it retrieves information.
* If the system returns irrelevant or low-quality data, its performance is poor.
* Evaluation involves determining merit, worth, and significance using specific criteria.

Key Evaluation Matrics :-

- Precision :- How many of the retrieved results are actually relevant?
- Recall :- How many of the total relevant Results were successfully retrieved?

EX :- Suppose an IR System retrieves 100 results, but only 80 are relevant →

~~Pre~~ precision = 80%.

However, if there were actually 200 relevant results available, but the system retrieved only 80, then Recall = 40%.

* The system should improve both metrics for better performance.
* The efficiency and accuracy of the system must be tested regularly.

→ LANGUAGE MODELLING

Introduction :-

* A language model in NLP is a probabilistic statistical model that determines the probability of a given sequence of words occurring in a sentence based on the previous word. It helps to predict which word is more likely to appear next in the sentence.
* There are 2 approaches for language modeling:-

1) Various Grammer-based model:-
   It uses predefined rules and structure of grammer.
   Ex:- "She is reading a book".

2) Statistical language model:-
   It uses probabilities derived from large data sets to predict the next word.